NAO Robot Limb Control Method Based on Motor Imagery EEG

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***Abstract*—Brain computer interface (BCI) technology refers to the technology that helps establish direct communication and control channels between the human brain and the computer. This technology can meet the control requirements of external equipment without passing through the patient's damaged nerves and declining muscle system. Electroencephalogram (EEG) contains a variety of information, including motor imagination information, steady-state visually evoked EEG information, emotional information, and fatigue information, etc. This paper designs a brain-computer interface system based on motor imagination and uses it to control the body movements of the NAO robot. The system is mainly divided into three modules: signal acquisition, signal processing and robot control. In terms of signal acquisition, electrode caps, conductive glue, EEG amplifiers and Curry7 software are used. In this paper, we design a frequency band feature selection algorithm based on an improved sub-band filter bank (FB)-Common Space Pattern (CSP). These features are classified using Bagging ensemble classification algorithm and converted to use Commands for controlling the NAO robot. In terms of communication control, the instruction codes are preset, and the classification results are corresponding to instructions for controlling different actions of the robot. It is transmitted to the robot by using wireless communication technology. The Bagging ensemble classification method proposed in this paper classifies EEG features. The accuracy of the four types of motor imaging EEG classification is as high as 78.29%, and the accuracy of NAO robot action execution experiments is up to 86.33%, which proves the correctness of the algorithm studied in this paper. The reliability and feasibility of the NAO robot brain control system designed in this paper.**

***Keywords*—Brain-computer interface, EEG, subband filter, co-space algorithm, integrated learning algorithm, motion imagination, NAO robot**

I. Introduction

The characteristic of "gradual freezing syndrome" patients is that the nervous system is not damaged after the illness, the brain consciousness and memory will not decline, but the motor nerve center in the spinal cord declines, muscle atrophy, the limbs move slowly and finally cannot move. Because the body's limbs are unable to move, as if freezing slowly, patients with "gradual freezing syndrome" lose the ability to take care of themselves [1].

This article aims to study the analysis of EEG signals of multiple types of motor imagination in BCI system [2]. By studying the preprocessing and feature extraction methods of EEG signals, the EEG signal features of motor imagination are extracted and screened to obtain EEG feature vectors that can represent different types of motor imagination. Finally, the classification method is studied, and it is hoped that a classifier suitable for EEG features can be obtained to achieve a good classification effect of motor imaging EEG features. Finally, it is applied to the NAO robot control system to realize the function of motor imagination EEG to control the body movements of the NAO robot.

II. System Design

The NAO robot brain control system based on the brain-computer interface is the practical application of the motor imagery EEG signal preprocessing, feature extraction and classification algorithms. The brain control system is mainly composed of five modules: brain electrical signal acquisition module, brain electrical signal analysis module, control instruction conversion module, control signal transmission module, and control equipment module [3]. The overall block diagram of the system structure design is shown in Fig. 1.

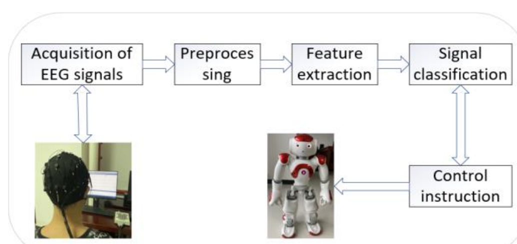

Fig. 1.Overall block diagram of NAO robot brain control system design

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III. Data collection and processing

A. Acquisition of EEG signals

In the EEG collection of this article, we choose the NuAmps series EEG collection equipment of Neuroscan Company in the United States as the collection equipment of the laboratory motor imagination EEG. The collection electrode cap used in the experiment is the Quik-Cap electrode cap designed by Neuroscan company, which conforms to the international 10-20 system standard [4]. The number of leads can be selected according to the experimental needs. After the subject wears the electrode cap, the researcher needs to inject conductive paste into the round hole of the electrode cap to reduce the impedance between the electrode and the scalp. In general, the electrode impedance is required to be less than 5kΩ. After injecting GND (ground terminal) and REF (reference terminal), the Curry7 software supporting the Neuroscan system can display the resistance values of other electrodes in real time. When injecting the conductive paste, remove the hair and rotate the syringe to inject the conductive paste into the electrode hole so that the conductive paste fills the entire electrode cavity. Fig. 2 shows an EEG signal acquisition diagram.

Fig. 2. Motor imagination EEG data acquisition experiment diagram

The EEG signals in this article only need to obtain the motor imaginary EEG signals of the three electrodes. Therefore, this experiment only needs to collect the EEG signals on the three leads C3, C4, and Cz. In addition, the ocular signals are collected as Remove artifacts. These three leads are located in the subject's occipital lobe, that is, the motor sensory cortex area, which can well reflect the subject's imagination and thinking signals [5-6].

B. Pretretment of EEG signals

EEG signals have strong nonlinear, non-stationary, and time-varying characteristics. EEG signal potential changes in small amplitude, usually in picovolts, so it is susceptible to interference [7]. There are two main sources of EEG signal noise, one is the artifact interference caused by the subjects' physiological activities, such as ocular artifacts, and the other is the artifact interference caused by power frequency noise [8]. Therefore, it is of great significance to denoise the EEG signal before the EEG signal feature extraction. We use the Butterworth filter to intercept the EEG signal in the frequency range, and propose an improved sub-band pass filter bank (Filter Bank, FB) filtering method based on the traditional Butterworth filtering method.

The mathematical definition of Butterworth filter is:

$$|H(j\omega)|^2 = 1/\left(1+\left(\frac{\omega}{\omega_c}\right)^{2n}\right) \quad (1)$$

where, ω_c is the cutoff frequency of the passband, and n is the filter order of the Butterworth filter [9].

C. Feature extraction method

The sub-band pass filter bank filters the same time-domain signal to obtain multiple sets of frequency-domain information of the sub-pass band. Combined with Common Spatial Pattern (CSP) filtering, the difference between the four types of motion imaging frequency band characteristics is expanded [10]. Finally, normalize and standardize the EEG data to obtain dimensionless EEG frequency domain feature data. We further select the frequency band features by using Principal Component Analysis (PCA) and the minimum absolute contraction selection operator (Least Absolute Shrinkage and Selection Operator (Lasso) reduces the dimensionality of the EEG feature vector and removes the linearly related EEG features [11-12]. Finally, the EEG feature FB-CSP data and the dimensionality reduction feature data are fused [13].

The commonly used normalization method is to map the data to [0,1] by linear transformation of the original data. The transformation function is:

$$x' = \frac{x-min}{max-min} \quad (2)$$

where, *min* is the minimum value in the sample, and *max* is the maximum value in the sample [14].

The commonly used standardization method is z-score standardization. The mean value of processed data is *0* and the standard deviation is *1*. The method is as follows:

$$x' = \frac{x-\mu}{\sigma} \quad (3)$$

where, μ is the mean value of the sample, and σ is the standard deviation of the sample. They can be estimated from the existing samples [15].

D. Signal classification

Before classification, we first build an integrated learning framework. Ensemble Learning (EL) is to establish multiple classification models on a data set, and integrate the classification results of multiple models to obtain the final classification results [16]. In this experiment, K-nearest neighbor (KNN), decision tree (DT) and twin support vector machine (TWSVM) are selected as the base estimator (Base Estimator), the integrated framework adopts the Bagging integrated learning framework [17], combined with the strategy selection Bayesian Sum Rule BSR [18].

K-Nearest Neighbor (KNN) is a machine learning algorithm that can be used for both classification and regression. The working principle of KNN is to find the closest K sample data in the data based on the distance metric, and then use the location information of the K closest "neighbors" to predict the category of the sample point [19].

Decision Tree (Decision Tree) is based on the known probability of occurrence of various situations, by constructing a decision tree to obtain the probability that the expected value of the net present value is greater than or equal to zero, evaluate project risk, and determine its feasibility. A graphical method of intuitive use of probability analysis [20].

Support Vector Machine (SVM) is a commonly used data classification and regression tool. The basic idea of support vector machine classification is to find the optimal hyperplane on the sample data of two different categories to realize the classification of samples of two different categories [21].

Twin Support Vector Machines (TWSVM) is an improved support vector machine classification algorithm [22]. TWSVM is essentially different from SVM. TWSVM is mainly for binary classification problems. SVM faces all classification problems. TWSVM is based on two sets of quadratic programming (QP), while SVM is based on a set of large quadratic programming [23].

IV. EXPERIMENT AND RESULTS DISCUSSION

A. Experiment design

The experiment designed in this paper is based on four types of motor imagination tasks: left hand, right hand, leg and tongue. The subjects are 6 students in the laboratory, and the 6 subjects are all healthy boys, aged 20-25 They are all ages, have no mental disorders, are emotionally stable, move their limbs normally, and meet the requirements for intelligence. The motor imagination EEG data sets collected in the laboratory are numbered T01, T02...T06. The subjects are required to perform corresponding motion imagination actions under the guidance of the arrow on the screen, and the EEG acquisition device collects motor imagination EEG data. The specific process of the experiment is shown in Fig. 3.

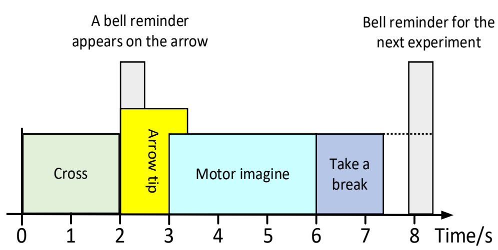

Fig. 3. Specific flow chart of a single experiment

In the arrow prompt mark, the left arrow represents the execution of left-hand motor imagination action, the right arrow represents the execution of right-hand motor imagination action, the up arrow represents the execution of tongue motor imagination action, and the down arrow represents the execution of foot motor imagination action.

B. NAO Robot Contro

We use the widely used wireless LAN technology to connect the robot to the wireless LAN. At the same time, the above-mentioned EEG acquisition and processing equipment is also located in the same LAN. The processing result is sent to the terminal device in real time. Through the preset instruction translation, real-time interactive effects can be achieved [24]. We use left-hand, right-hand, leg, and tongue as the experimental paradigms, which correspond to the tasks of raising the left hand, raising the right hand, squatting and standing of the robot.

In the control process, the command sign corresponding to the real-time motion image classification result is sent to the server via the wireless network. Each time the server receives the status value, it will receive the data to create a TXT file, and write the status value into the TXT file [25].

In the experiment, the robot is controlled by the Python API of the robot control software, and the script reads the newly written txt file in the specified folder. When the command symbol reads "00", it will control the robot to perform no operations. When the command symbols are "11", "22", "33" and "44", control the robot to raise the left hand, raise the right hand, squat and stand up respectively.

C. Result and discussion

The ensemble learning classification framework constructed in this article is used to classify the motor imagination EEG data of six subjects in the laboratory, and the classification accuracy of each subject's EEG data set in each base evaluator is recorded separately, and based on BSR Comparison of the classification accuracy of the Bagging integrated classification algorithm (EL). The classification results are shown in Table 1.

Table1 Classification results of laboratory motor imaging EEG(%)

Data Set	KNN	DT	TWSVM	Average	EL
T01	65.43	63.2	71.14	66.62	75.71
T02	64.29	62.1	68.52	64.98	74.57
T03	68.71	64.8	73.86	69.14	78.29
T04	70.29	69.7	71.91	70.64	76.29
T05	63.43	61.2	64.29	63.00	72.86
T06	68.62	63.4	73.34	68.46	78.14
Average	66.80	64.1	70.51	67.14	75.98

From the comparison of the data in Table 1, the accuracy of each test data set obtained by the ensemble learning classification algorithm has been improved, and the classification accuracy is the highest among the classification accuracy results obtained by the four classifiers. Compared with TWSVM, which has the best classification effect among all base evaluators, the accuracy rate is improved by 4%~6%, which verifies the effectiveness of the feature extraction method and classification method proposed in this paper.

In the test experiment, the results of data classification are converted into control instructions to control the NAO robot. Each action test is performed 100 times, and each action performed by the NAO robot is recorded. The response results are shown in Table 2, and the relationship between the motor imagery accuracy of different testers and the task category is shown in Fig. 4.

Table2 Motion imaging NAO robot action accuracy (%)

Testors	Raise left hand	Raise right hand	Squat	Stand up	No action
T01	76	74	64	59	86
T02	73	76	61	57	83
T03	79	71	67	62	91
T04	77	68	65	61	89
T05	69	71	59	52	83
T06	78	75	66	62	86
Average	75.33	72.50	63.67	58.83	86.33

By comparing the results of NAO robot response actions, it can be seen that the classification result of left-hand motor imagination has the highest accuracy, which is 75.33%. The right hand comes next with 72.50%. The accuracy of leg and tongue movement imaging was slightly lower, being 63.67% and 58.53% respectively. It shows that the classification accuracy of left and right hand motor imagery EEG signals is significantly higher than that of leg and tongue motor imagery EEG signals. When not performing specific motion imaging tasks, the accuracy of the NAO robot's inaction is 86.33%, which shows that the NAO robot brain control system based on brain-computer interface technology has a good effect of preventing misoperation.

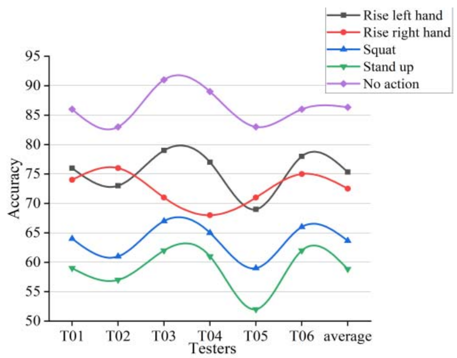

Fig. 4. The curve of different testers' motor imagination accuracy and task category

V. CONCLUSION

Based on the background of rehabilitation robots and the human-machine interface based on motion imagination, a complete BCI system was constructed to realize the control of the limbs of the NAO robot, and the motor imaging EEG classification and the action execution effect of the NAO robot were tested and verified. The effectiveness of the frequency band feature selection algorithm and classification method proposed in this paper is discussed. Collect 6 students' motor imagination EEG signals and use the Bagging ensemble classification method proposed in this article to classify EEG features. The four types of motor imagery EEG classification accuracy rate are up to 78.29%, and the NAO robot action execution experiment has the highest accuracy rate. It reaches 86.33%, which proves the correctness of the algorithm studied in this paper and the reliability and feasibility of the NAO robot control system designed in this paper.

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